$$\boxed{\int_a^b x^p\,dx = \frac{x^{p+1}}{p+1}\Big|_a^b ,\ p\in Z^+}$$

1. $b^p-a^p=(b-a)(b^{p-1}+b^{p-2}a+b^{p-3}a^2+\cdots+ba^{p-2}+a^{p-1})$, $p\in Z^+$

proof: right $= b(\ )-a(\ )$

$= b^p+b^{p-1}a+b^{p-2}a^2+\cdots+b^2a^{p-2}+ba^{p-1}$

$-(b^{p-1}a+b^{p-2}a^2+b^{p-3}a^3+\cdots+ba^{p-1}+a^p)$

$= b^p-a^p$

2. $n^p < \dfrac{(n+1)^{p+1}-n^{p+1}}{p+1} < (n+1)^p$, $p,n\in Z^+$

proof: $\because (n+1)^{p+1}-n^{p+1}=(n+1)^p+(n+1)^{p-1}n+\cdots+n^p$
$=: A$ (from 1)

$\therefore (p+1)n^p < A < (p+1)(n+1)^p$

$\therefore n^p < \dfrac{A}{p+1} = \dfrac{(n+1)^{p+1}-n^{p+1}}{p+1} < (n+1)^p$

3. $\sum_{k=1}^{n-1} k^p < \dfrac{n^{p+1}}{p+1} < \sum_{k=1}^{n} k^p$, $n,p\in Z^+$

proof: let $A(n)$ denote the above statement.

$A(1)$ is true: $0<\frac{1}{p+1}<1$

if $A(n)$ is true,

then, $\because \sum_{k=1}^{n-1} k^p < \dfrac{n^{p+1}}{p+1} < \sum_{k=1}^{n} k^p$ ($A(n)$ true)

与 $\because n^p < \dfrac{(n+1)^{p+1}-n^{p+1}}{p+1} < (n+1)^p$ (from 2)

add them,

$\therefore \sum_{k=1}^{n} k^p < \dfrac{(n+1)^{p+1}}{p+1} < \sum_{k=1}^{n+1} k^p$

$\therefore A(n+1)$ is true. ∎

4. $\int_0^b x^p\,dx$, $b>0$, $p\in Z^+$

solve: ① $\because x^p$ bounded and $\uparrow$ on $[0,b]$.

$\therefore \int_0^b x^p\,dx\ \exists$.

② $\because \sum_{k=1}^{n-1} k^p < \dfrac{n^{p+1}}{p+1} < \sum_{k=1}^{n} k^p$, $n,p\in Z^+$ (from 3)

multiply by $\dfrac{b^{p+1}}{n^{p+1}}$

$\therefore \dfrac{b}{n}\sum_{k=0}^{n-1}\left(\dfrac{kb}{n}\right)^p < \dfrac{b^{p+1}}{p+1} < \dfrac{b}{n}\sum_{k=1}^{n}\left(\dfrac{kb}{n}\right)^p$

let $f(x)=x^p$,

$\therefore \dfrac{b}{n}\sum_{k=0}^{n-1} f(x_k) < \dfrac{b^{p+1}}{p+1} < \dfrac{b}{n}\sum_{k=1}^{n} f(x_k)$

$\therefore \int_0^b x^p\,dx = \dfrac{b^{p+1}}{p+1}$ (by theorem 1.13 of Apostol's Calculus, Volume 1)

5. $\int_0^b x^p\,dx = \dfrac{b^{p+1}}{p+1}$, $b<0$, $p\in Z^+$

proof: $\because \int_0^b f(x)\,dx = -\int_0^{-b} f(-x)\,dx$
(reflection property)

$\therefore \int_0^b x^p\,dx = -\int_0^{-b}(-x)^p\,dx$

$=(-1)^{p+1}\int_0^{-b} x^p\,dx$

$=(-1)^{p+1}\dfrac{(-b)^{p+1}}{p+1}$ (from 4)

$=\dfrac{b^{p+1}}{p+1}$

6. $\int_0^b x^p\,dx = \dfrac{b^{p+1}}{p+1}$, $\forall b\in R$, $p\in Z^+$

7. $\int_a^b x^p\,dx = \dfrac{x^{p+1}}{p+1}\Big|_a^b$, $a,b\in R$, $p\in Z^+$

proof: $\int_a^b x^p\,dx = \int_0^b x^p\,dx - \int_0^a x^p\,dx$
(additive property)

$= \dfrac{b^{p+1}-a^{p+1}}{p+1}$

$= \dfrac{x^{p+1}}{p+1}\Big|_a^b$